

Scalability & Future Plan

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Interaction history lives only in Streamlit session state and is lost on refresh; CSV logging is append-only with no query layer.	Learners cannot see long-term trends across multiple visits.	High
2	Single-user, no authentication — anyone using the local instance shares the same session.	Not viable for multiple concurrent learners without change.	High
3	Synchronous inference (BiLSTM + BERT + Gemini all run in the request path) can make the UI feel slow under load.	Perceived latency increases as models and traffic grow.	Medium
4	No automated tests around the prediction pipeline.	Regressions could go unnoticed when the models or code change.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user, in-memory session state only	Move to a hosted deployment (e.g. Streamlit Community Cloud) with per-user session isolation
2	Data Storage	Flat CSV file (emotion_response_examples.csv)	Migrate to a lightweight database (SQLite → PostgreSQL) for querying and multi-user history
3	Performance	Synchronous BiLSTM + BERT + Gemini calls per request	Cache model objects (already done at import time), consider async/batched inference for concurrent users
4	Security	.env-based API key management only, no auth layer	Add user authentication and move secrets to a managed secret store for a hosted deployment

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add a README, .env.example, and fix the CSV filename mismatch bug	[Add date]	Improves onboarding and evaluator experience

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Persist history in a lightweight database and add basic user identification	[Add date]	Enables trend tracking across sessions
Phase 3	Deploy publicly (e.g. Streamlit Community Cloud) with automated tests in CI	[Add date]	Makes the app accessible beyond local demos and safer to change
Phase 4	Extend emotion coverage to more languages and add voice/text-message input	[Add date]	Widens the learner base the platform can support